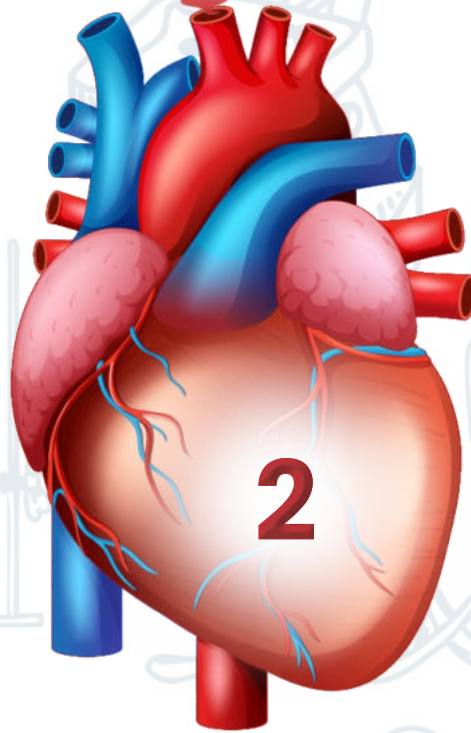
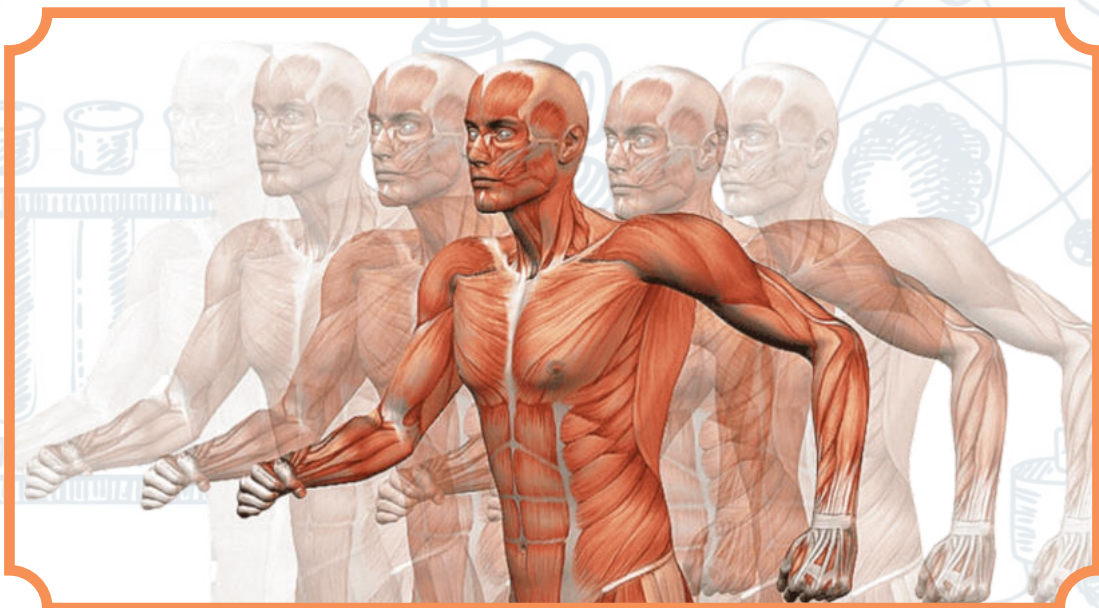


UNIT



ENZYMES



ENZYMES(BIO-CATALYSTS)

Enzymes are bio-catalyst which speed up the chemical reactions by lowering Energy of activation

ENERGY OF ACTIVATION

Amount of energy which is required to start a chemical reaction. OR Energy required to break a (particular covalent) bond present in reactant.

NOMENCLATURE OF ENZYMES

Enzyme is a Greek word means-En(in) and Zyme(yeast).

NATURE OF ENZYME

Almost all enzymes are protein in nature except few which are nitrogenous acids like RNA-DNA(Ribozymes). Ribozymes catalyze reactions in genetic informations.

CHARACTERISTICS OF ENZYMES

- Protein in nature and are formed by living cells.
- May be intracellular or extra cellular.
- Remains unchanged during and after the reaction.
- Speed up the rate of reaction by decreasing energy of action.
- Specific in their nature.
- Heat sensitive and act on particular (optimum) temp.
- Each has specific substrate pH for its activity.
- Action can be altering by activators and inhibitors.

CLASSIFICATION OF ENZYME (ON THE BASIS OF STRUCTURE)

Enzyme consist of only protein (e.g.Amylase and Pepsin) Conjugated or Holoenzymes: May contain a non-protein part "Prosthetic group" as well (e.g. Phosphatase and Peptidase)

Holoenzyme = Apoenzyme + Prosthetic group
(Protein part) (Non-protein part)

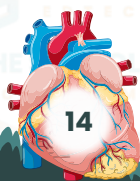
CLASSIFICATION OF ENZYME (ON THE BASIS OF FUNCTIONS)**(1) OXIDOREDUCTASE**

Catalyze reactions in which one substrate is oxidized while other is reduced.
Sub classes are:

- Dehydrogenases(convert single bond to double bond)
- Oxidases (use oxygen as oxidant)
- Peroxidases (use H_2O_2 as oxidant) Hydroxylases (introduce hydroxyl group)
- Oxygenases (introduce mol. Oxygen in place of double bond).

(2) TRANSFERASES

Transfer one carbon group (e.g. methyl) from one substrate to another substrate.



(3) HYDROLASES

Catalyze hydrolytic cleavage of C-O, C-N, C-C and P-O bonds and other single bonds (e.g. Peptidases, Esterases, Glycosidases and Phosphatidases).

(4) LYASES

Catalyze Elimination reactions to form double bond and reversible reaction by adding groups across double bond (e.g. Decarboxylases, Aldolases and Dehydratases).

(5) ISOMERASES

They alter the structure but not the atomic composition by moving a group from one position to another in one molecule (e.g. Epimerases, Mutases).

(6) LIGASES

Catalyze reaction in which two molecules are joined. They are also known as synthetases.

MODE OF ACTION OF ENZYMES

Following two hypotheses explain the mode of enzyme action.

1- Lock and Key Hypothesis

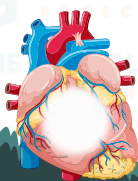
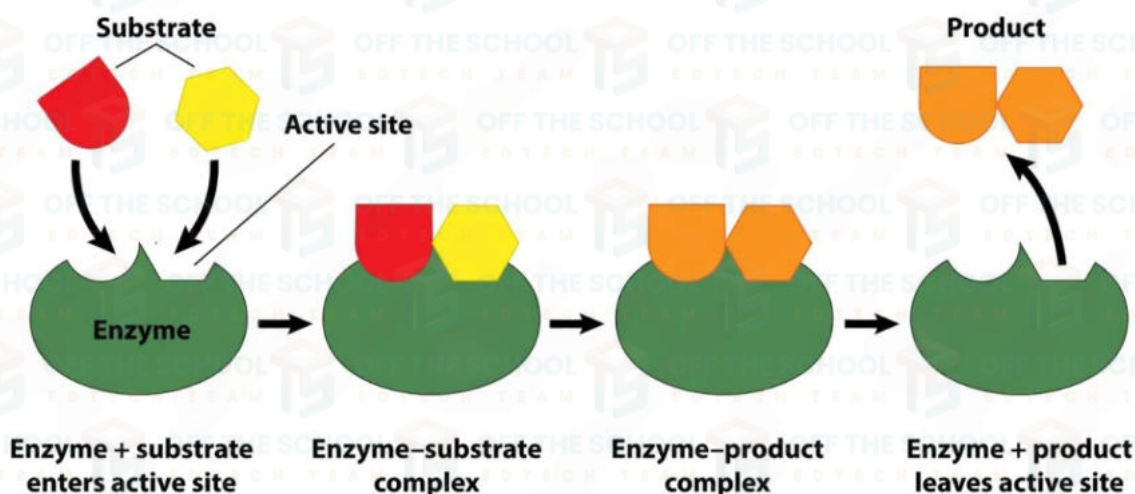
2- Induced fit hypothesis

LOCK AND KEY MODEL

Fischer in 1890 suggested that each enzyme had a particular shape into which the substrate fits exactly. This was known as the lock and key hypothesis. According to this hypothesis, the substrate is imagined like a lock while the enzyme is imagined like a key.

As one specific key can open only a specific lock, similarly a specific enzyme can break up only one specific substrate. The active site is regarded as a rigid structure that does not modify or change during the reaction process. However, later studies did not support this hypothesis in all type of reactions and therefore the hypothesis was modified into Induced fit hypothesis.

Lock and Key Model



INDUCE FIT MODEL

The attraction of the substrate and enzyme form an enzyme-substrate complex. It was originally referred to as the Lock and Key Enzyme Theory. The current theory suggests that the enzyme molecules are in an inactive form. To become active they must undergo a slight change in structure to form.

Process of enzyme catalysis:

There are three ways of thinking about enzyme catalysis which are:

- i- Reaction mechanism
- ii- Molecular geometry
- iii- Energy changes

i- Reaction mechanism:

In any chemical reaction, a substrate (S) is converted into a product (P):



There may be more than one substrate and more than one product, but that doesn't matter. In an enzyme-catalyzed reaction, the substrate first binds to the active site of the enzyme to form an enzyme-substrate (ES) complex, then the substrate is converted into product while attached to the enzyme, and finally the product is released.

ii. Molecule Geometry:

The substrate molecule fits into the active site of the enzyme molecule like a key fitting into a lock. Once there, the enzyme changes shape slightly, distorting the molecule in the active site, and making it more likely to change into the product.

Example:

If a bond in the substrate is to be broken, that bond might be stretched by the enzyme, making it more likely to break. Alternatively, the enzyme can make the local conditions inside the active site quite different from that outside (such as pH, water concentration, charge), so that the reaction is more likely to happen.

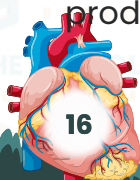
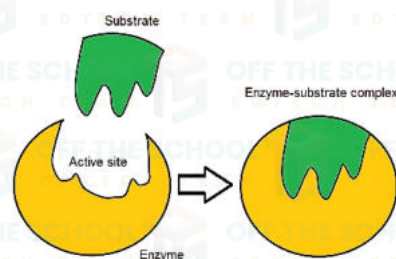
iii- Energy changes:

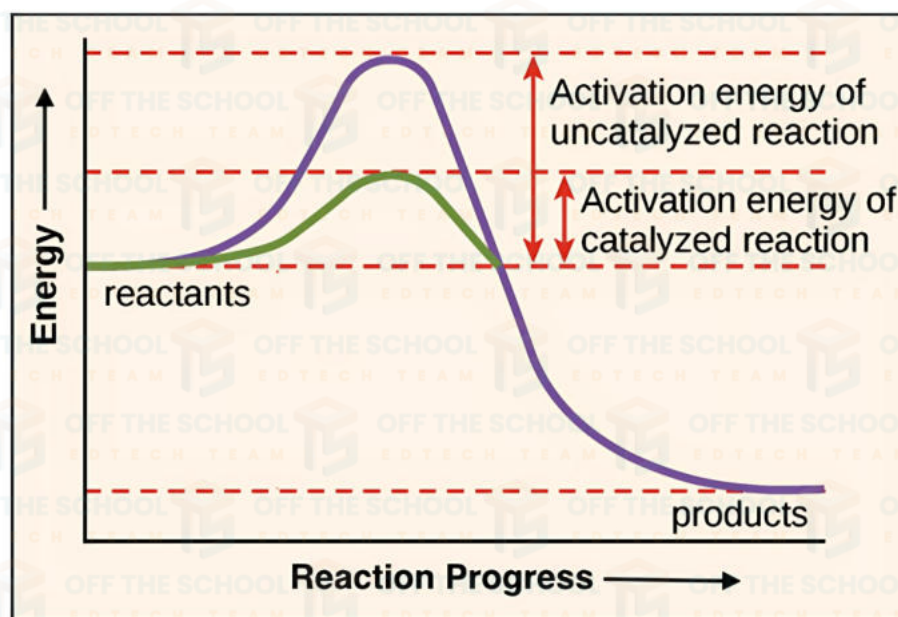
The way enzymes work can also be shown by considering the energy changes that take place during a chemical reaction. Consider a reaction where the product has lower energy than the substrate, so the substrate naturally turns into a product (in other words the equilibrium lies in the direction of the product).

ACTIVATION ENERGY:

Before changing into a product, the substrate must overcome an "energy barrier" called the activation energy (EA). The larger the activation energy, the slower the reaction will be because only a few substrate molecules will by chance have sufficient energy to overcome the activation energy barrier.

Enzymes dramatically reduce the activation energy of a reaction, so that most molecules can easily get over the activation energy barrier and quickly turn into a product.





ENZYME INHIBITION

Inhibitors:

"An enzyme inhibitor is a molecule that binds to an enzyme and decreases its activity."

Inhibitors inhibit the action of enzymes, reducing the rate of their responses. They are found inherently but are also used artificially as drugs, pesticides and analysis tools.

Types of inhibitors:

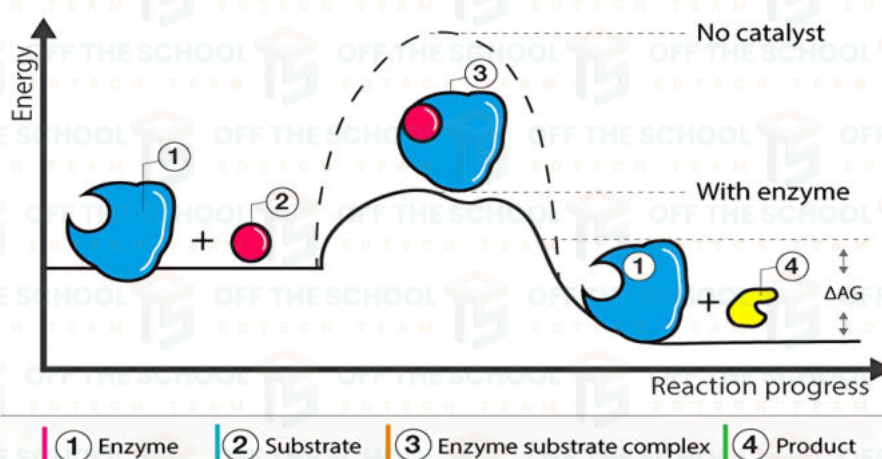
There are three kinds of inhibitors.

1. Competitive Inhibitor
2. Non-Competitive Inhibitor
3. Un-Competitive Inhibitor

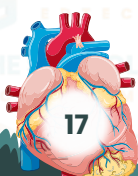
Competitive Inhibition

A molecule other than the substrate binds to the enzyme's active site, causing competitive inhibition. The inhibitor (molecule) has a structural and chemical similarity to the substrate (hence able to bind to the active site). The competitive inhibitor hinders substrate binding by blocking the active site. Since the inhibitor competes with the substrate, increasing the substrate concentration reduces the inhibitor's actions.

ACTION AND NATURE OF ENZYMES

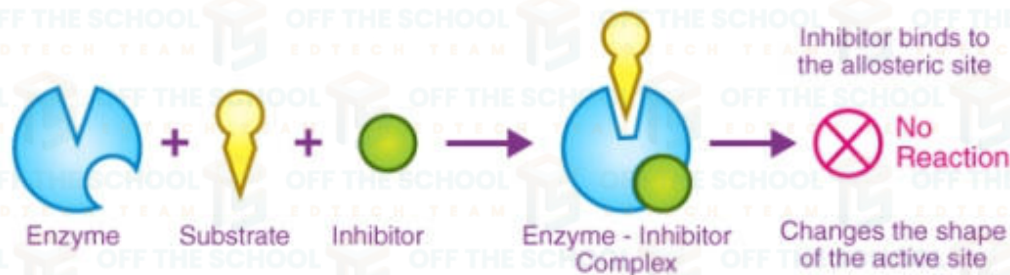


① Enzyme ② Substrate ③ Enzyme substrate complex ④ Product



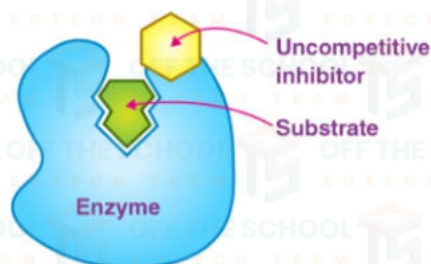
2. Non-Competitive Inhibition

A chemical binds to a location other than the active site in non-competitive inhibition (allosteric site). When the inhibitor binds to the allosteric site, the enzyme's active site undergoes a structural shift. The active site and substrate no longer share affinity as a result of this alteration, preventing the substrate from binding. Increased substrate levels will not be able to reverse the inhibitor's action since the inhibitor is not in direct competition with the substrate.



3. Uncompetitive Inhibition

The inhibitor binds only to the substrate-enzyme complex in uncompetitive inhibition. In reactions involving two or more substrates or products, uncompetitive inhibition is common. Non-competitive inhibition can occur with or without the presence of the substrate, whereas uncompetitive inhibition requires the formation of an enzyme-substrate complex.



Feedback Inhibition:

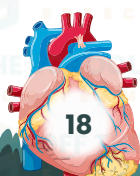
In feedback inhibition, there is a second binding site on the enzyme where the inhibitor binds so that the inhibitor is not necessarily similar in structure to the substrate. The absence or presence of the inhibitor at this second binding site activates or deactivates the enzyme, by changing the conformation of the enzyme so that the active site is made available or unavailable to the substrate. The inhibitor is usually the product of a reaction formed during the metabolic pathway.

FACTORS AFFECTING ENZYME ACTIVITY

The activity of an enzyme can be affected by its environmental factors. Temperature, pH, substrate concentration, and modulators are essential parameters that influence the rate of enzyme-catalysed reactions.

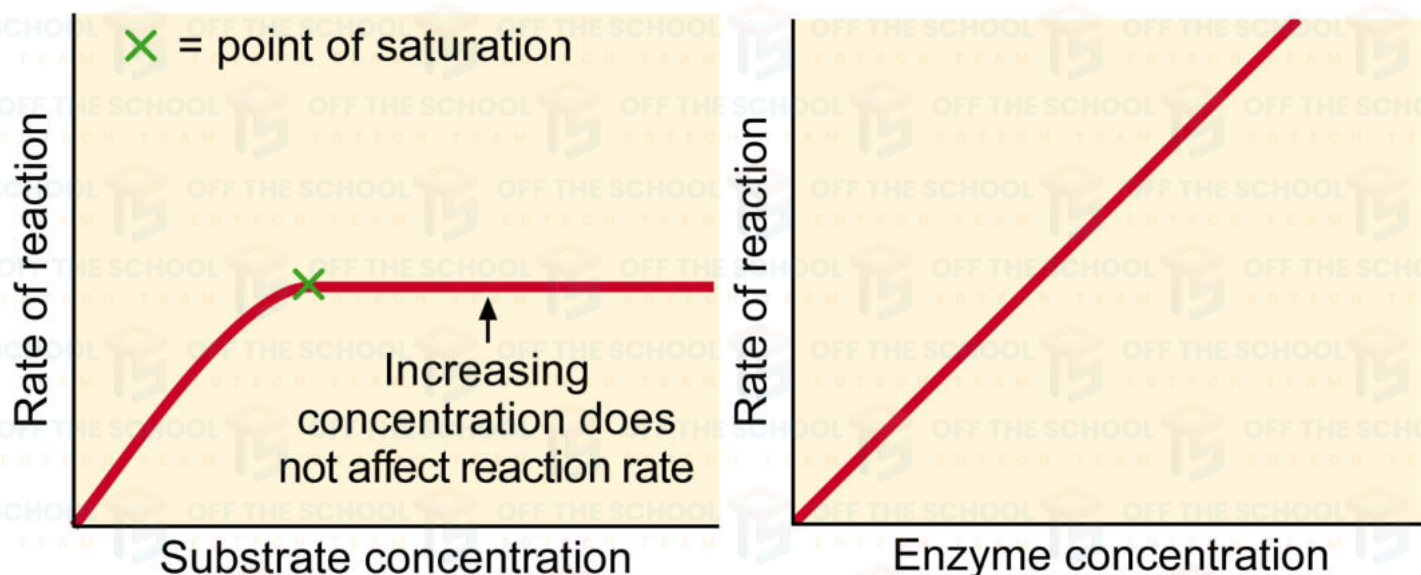
1. SUBSTRATE CONCENTRATION

The rate of an enzyme-catalyzed reaction shows a curved dependence on substrate concentration. As the substrate concentration increases, the rate increases because more substrate molecules can collide with enzyme molecules, so more reactions will take place. At higher concentrations, the enzyme molecules become saturated with the substrate so there are few free enzyme molecules so adding more substrate doesn't make much difference.



2. ENZYME CONCENTRATION

As the enzyme concentration increases the rate of the reaction increases linearly because there are more enzyme molecules available to catalyze the reaction. At very high enzyme concentration the substrate concentration may become rate-limiting, so the rate stops increasing. Normally enzymes are present in cells in rather low concentrations.



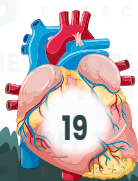
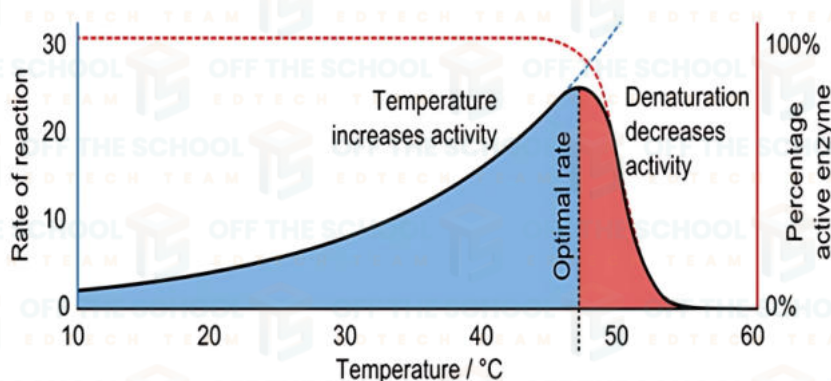
3. TEMPERATURE

An optimum temperature range is required to maximize the enzyme activity. Temperatures that are either higher or lower than the optimal temperature causes a decrease in the enzyme activity and can lead to denaturation (when the temperature is too high).

The temperatures between 37 and 45 are referred to as the optimal temperature for most enzymes. At this given temperature, most enzymes become more active. Extremely high temperatures can cause an enzyme to lose its denature form and eventually stop functioning. The majority of enzymes in the human body have a temperature optimum of 37°C and are denatured or destroyed at higher temperatures.

However, there are some exceptions, for example, in **extremophiles**. Few enzymes, such as Taq DNA polymerase found in **thermophilic bacteria**, are active at temperatures as;

Optimum Temperature of Enzyme

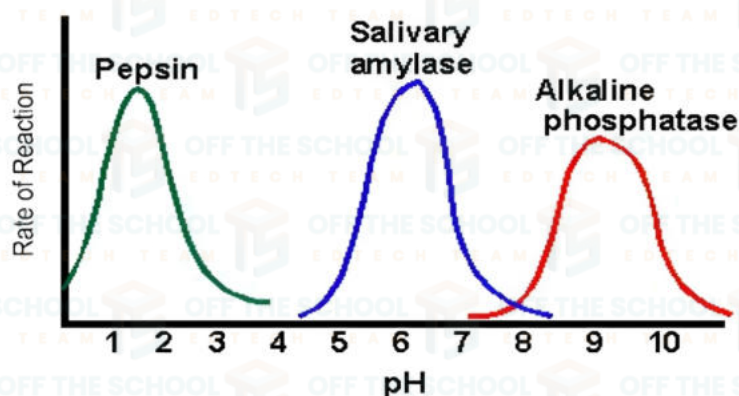


4. pH

Enzymes are **pH specific** i.e. work in specific pH (because of protein can act both in acidic and basic medium). pH has an impact on enzyme activity. A bell-shaped curve emerges when enzyme activity is plotted versus pH. Each enzyme has a certain optimal pH at which the reaction rate is the fastest. The optimal pH is when a specific enzyme's activity is at its peak. The enzyme activity is greatly reduced below and above the optimal pH, and at high pH, the enzyme becomes completely inactive. For most enzymes, the optimum pH for the enzymatic activity of ranges from 6 to 7.

Example

- The salivary **amylase** enzyme is most active at pH 6.8. The reaction rate decreases, and the enzymes denatured above and below this optimal pH range.
- Pepsin's optimal pH is 1.8 because the carboxylic acid group is protonated at low pH, which allows pepsin to catalyse the reaction.
- Alkaline phosphatase optimum pH is 9.



THINGS TO BE REMEMBER

INHIBITORS

Substances which decrease the activity of enzymes.

COMPETITIVE INHIBITORS

Inhibitor molecules which resemble the normal substrate molecule and compete for admission into the active site.

They block the substrate from entering active site.

NON-COMPETITIVE INHIBITORS

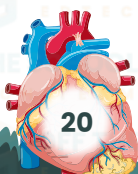
Inhibitors bind to a part of the enzymes away from the active site (Allosteric site). This binding causes a change in the enzyme molecule shape and a decrease in enzyme activity.

FEED BACK INHIBITION

Common biological control mechanism of brain in order to regulate enzyme activity.

PROSTHETIC GROUP

Non-protein part of enzyme (Co-enzyme or Co-factor)



CO-ENZYME

When prosthetic group consist of organic molecules (like FAD/NAD)

CO-FACTORS/ACTIVATORS

When prosthetic group consist of inorganic molecules (like Ca^{++} , Na^{+} etc).

APOENZYME

Protein part of enzyme.

